B.Sc. Engg. CSE 6th Semester 25 June 2026 (Morning)

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION SUMMER SEMESTER, 2024–2025

DURATION: 1 HOUR 30 MINUTES FULL MARKS: 50

CSE 4615: Wireless Networks

Programmable calculators are not allowed. Do not write anything on the question paper.
Answer all 3 (three) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

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1. a) Clarify the broadcast limitations at wireless links by answering the following questions: 2 - (C (P
   i. Why is an *explicit acknowledgement* from the receiving station to the transmitting station necessary in wireless communication, but typically not required in wired networks?
   ii. Why does the *hidden terminal problem* occur in multi-hop wireless networks, but not in wired networks?

   b) A wireless transmitter operates at $P_{tx} = 30$ dBm and the receiver sensitivity is $P_{rx} = -82$ dBm. 2 (C (I
   i. Calculate the maximum tolerable path loss in dB.
   ii. The measured received power at a particular distance is 1 μW. Determine whether the signal can be meaningfully extracted at that distance. Show all necessary calculations.

   c) In the context of *Rate Adaptation* used in 802.11 networks, explain how a base station and mobile device dynamically resolve the *maximising transmission rate vs. minimising probability of error* trade-off as the mobile moves away from the access point. Use the SNR-vs-BER curves for BPSK (1 Mbps), QAM-16 (4 Mbps), and QAM-256 (8 Mbps) in your explanation. (C (P

2. a) Briefly explain the difference between infrastructure-based and infrastructure-less wireless networks, giving one example of each. Why might an infrastructure-less network be preferable in an emergency or disaster relief scenario? 3 - (C (P

   b) Explain the *store-carry-and-forward* paradigm used in Delay Tolerant Networks (DTNs). How does this differ from the packet-switching model used in the conventional Internet? Under what circumstances does a DTN node store a bundle rather than forwarding it immediately? 2 - (C (P

   c) Vehicular Ad-hoc Networks (VANETs) face four major categories of challenges: *communication, networking, mobility*, and *security*. Choose any two of these categories and, for each, describe one specific challenge. Explain why these challenges are harder to solve in a vehicular environment than in a conventional wireless LAN. 2 - (C (I

3. a) IEEE 802.11 standard defines multiple Interframe Spaces (IFS) to provide differentiated access priority among frames. For 802.11a, the slot time is 9 μs, SIFS is 16 μs, and the contention window parameters are $CW_{min} = 16$ and $CW_{max} = 1024$. 2 - (C (P
   i. Calculate PIFS and DIFS for 802.11a, showing all calculations.
   ii. Explain why SIFS is shorter than DIFS and what practical consequence this ordering has on frame priority in the 802.11 DCF.

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b) The hidden terminal problem is a fundamental challenge in wireless networks. The Request to Send/Clear to Send (RTS/CTS) is commonly used to mitigate this problem. Explain how the RTS/CTS helps address the hidden terminal problem. Why does a hidden station refrain from transmitting after receiving a CTS frame, even though it did not receive the corresponding RTS frame? 5 (CO2) (PO2)

c) Consider a Basic Service Set (BSS) consisting of three stations: *Station A*, *Station B*, and *Station C*. All stations are within mutual transmission range. RTS/CTS is *enabled*, and $CW_{min} = 7$. 7 + 3 (CO2) (PO2)

Station A wishes to transmit an MAC Service Data Unit (MSDU) to Station C. Its *first transmission attempt fails* (assume no CTS is received after the RTS). After applying the appropriate backoff procedure, the *second transmission attempt succeeds*.

i. Draw a detailed timeline diagram (x-axis = time; y-axis = Station A, Station B/AP, Station C) showing all key events: DIFS intervals, backoff slot countdowns (with explicit counter values and CW range), RTS, CTS, DATA, and ACK frames, SIFS gaps, NAV periods for Station B, and the collision/failure event. Clearly annotate the moment of failure and the moment of success.

ii. Identify which station sets the NAV after the successful RTS/CTS exchange in the second attempt. Explain how the NAV timer protects the ongoing data exchange from interference.

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